Unit ID: 679b07ffaaf593b2823f304a

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738213372668-Chapter 6 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

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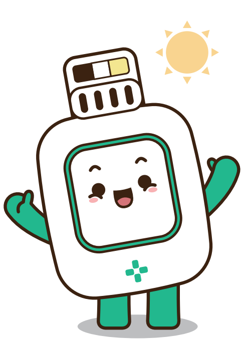
UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

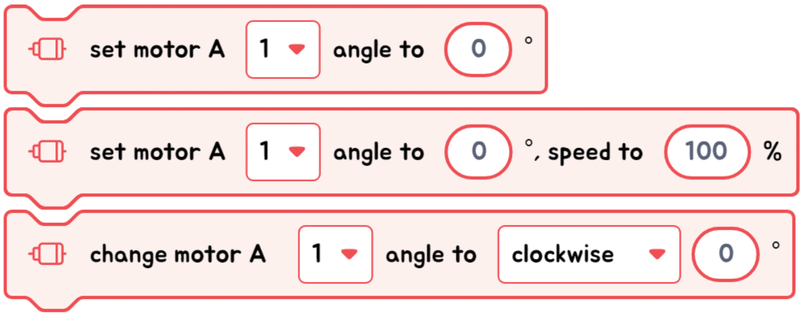
DUST-FREE CLOTHES: HOW CAN WE KEEP OUR CLOTHES CLEAN AFTER PLAYING OUTSIDE?

Alex wanted to go out and have fun but was warned on TV to stay indoors due to high acceptable dust levels. Despite the warning, Alex secretly decided to go out and play. On the way home, Alex felt anxious about a concern: what if she returned with dirt and dust on her clothes, which made her family sick? When Alex arrived home, she hoped the dust would disappear so she wouldn't have to worry about washing her clothes.

There is a growing trend to develop technologies to improve air quality and health. Air purifiers and AirDressers are representative examples. We use sensor robots to create an **AirDresser** that removes dust from clothes. This approach combines modern technology and practicality to make everyday life healthier.

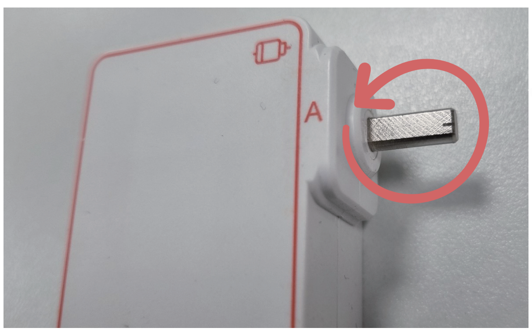


ABOUT MOTOR A MODULE

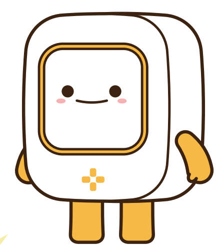


Motor A (same as B) can control speed and angle, and for angles, there is a block that allows you to specify the angle, the speed at which it moves by the angle, and the direction of rotation of the angle.

CODING ACCORDING TO ANGLES



Horn

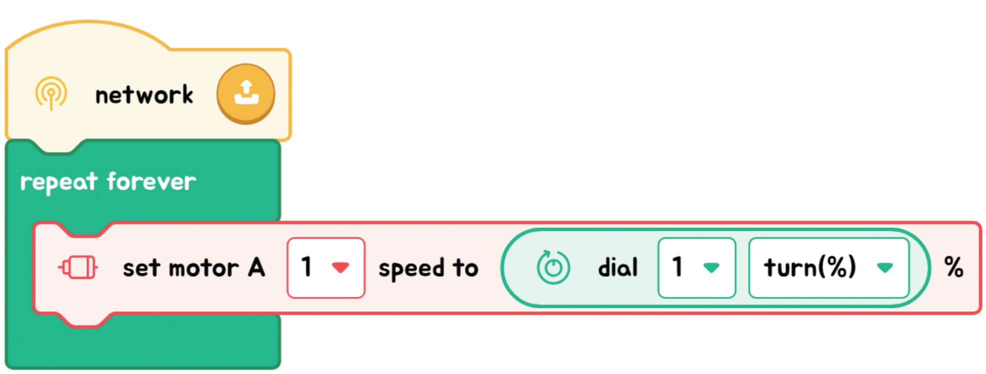


TIPS

- The grooved side of the motor shaft is the 0 point.
- Used to determine location when inserting the horn

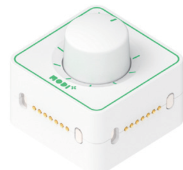
EXAMPLE OF SPEED-CONTROLLED CODING USING DIALS

Changing the coding value every time speed control is needed is inconvenient. What if we could adjust the speed with a dial to reduce this inconvenience? Let's write code to create a speed control with the same function as a radio's volume control.



Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?

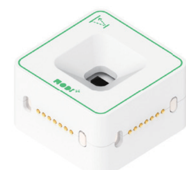


DIAL

BUTTON

ENVIRONMENT

JOYSTICK

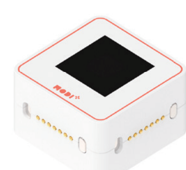


TOF

SPEAKER

IMU

LED

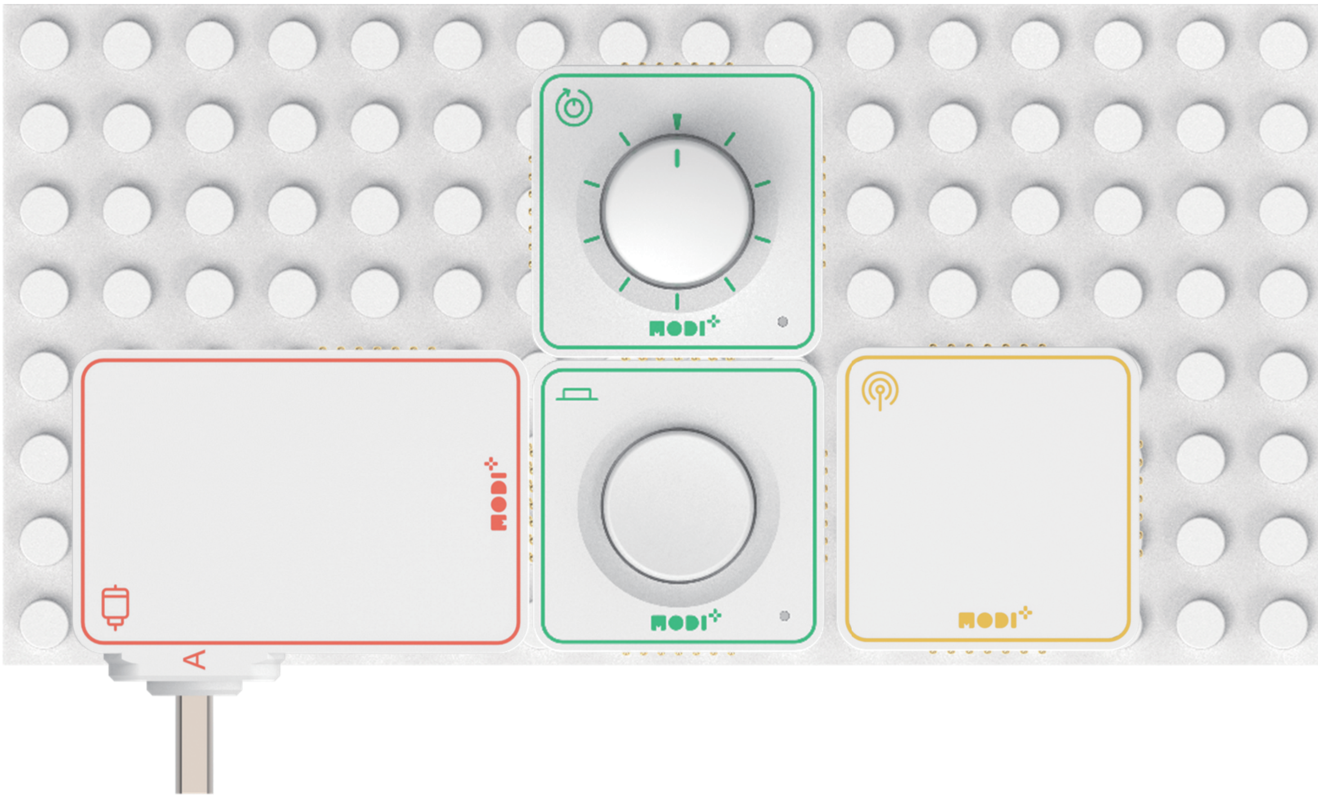


DISPLAY

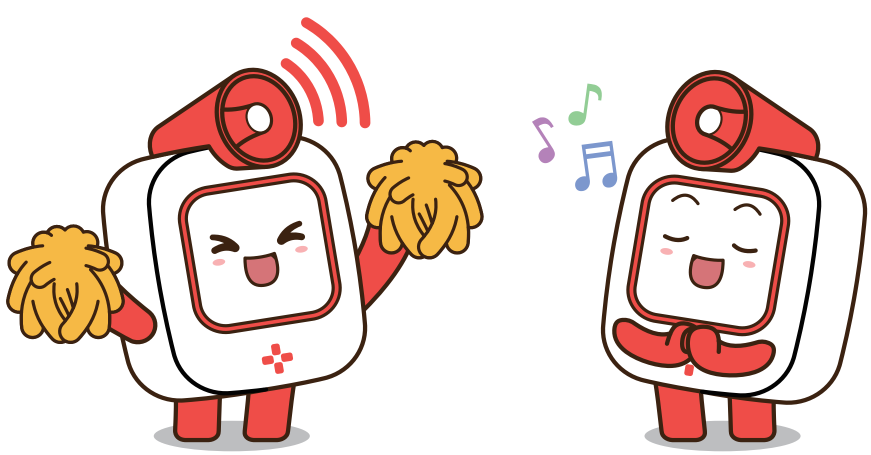
MOTOR A

MOTOR B

HOW DOES THE CREATION LOOK LIKE?

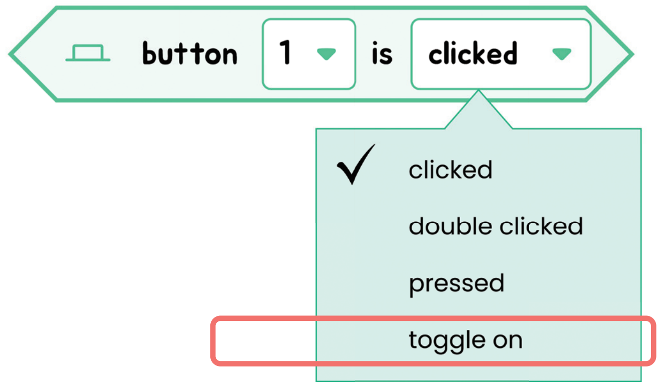


The AirDresser's shaking degree can be expressed as an angle; the angle is determined by monitoring, and the value is coded.



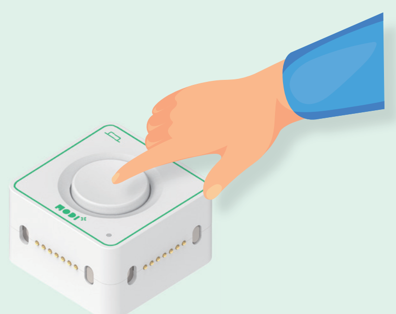
Module Functionality

Let's look detail about the Button module.



Button can receive 4 types of information.

1. Click

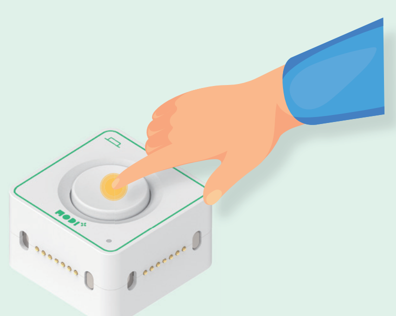


Detects if Button is clicked or not.
Range: True(Click once) : 100 / False : 0

2. Double click

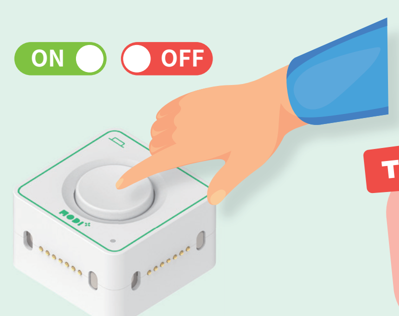
Detects if Button is double - clicked or not.
Range: True(Click twice) : 100 / False : 0

3. Press



Show the press status of Button.
Range: True(Pressed) : 100 / False : 0

4. Toggle



Changes on / off when Button is clicked.
Range: True(On) : 100 / False : 0